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SET-5(DSA0611-DH&V)

AIM

To analyze website traffic data using **R programming** and visualize page views, click-through rates, and user interactions using different charts.

ALGORITHM

1. Start RStudio.
2. Create the website traffic dataset containing Date, Page Views, and Click-through Rate.
3. Create a **line chart** to visualize daily page views.
4. Sort the data according to Click-through Rate and select the top 3 days.
5. Create a **bar chart** for the top 3 days with the highest Click-through Rate.
6. Create user interaction data containing Likes, Shares, and Comments.
7. Create a **stacked area chart** to visualize the user interactions.
8. Add suitable titles and axis labels.
9. Display the graphs.
10. Stop.

R CODE

```
website <- data.frame(  
  Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")),  
  PageViews=c(1500,1600,1400,1650,1800),  
  CTR=c(2.3,2.7,2.0,2.4,2.6)  
)
```

1. Line Chart

```
plot(website$Date,website$PageViews,  
     type="o",  
     main="Daily Page Views",  
     xlab="Date",  
     ylab="Page Views")
```

2. Bar Chart

```
top <- website[order(-website$CTR),][1:3,]
```

```

barplot(top$CTR,
        names.arg=as.character(top$Date),
        main="Top 3 Days by Click-Through Rate",
        xlab="Date",
        ylab="CTR (%)")

```

3. Stacked Area Chart

```

interactions <- cbind(
  Likes=c(100,120,90,130,150),
  Shares=c(50,60,40,70,80),
  Comments=c(30,40,25,45,50)
)

```

```

x <- 1:5

```

```

total <- rowSums(interactions)

```

```

plot(x,total,
     type="n",
     main="Website User Interactions",
     xlab="Date",
     ylab="Interactions",
     xaxt="n",
     ylim=c(0,max(total)))

```

```

axis(1,at=x,labels=format(website$Date,"%b %d"))

```

```

polygon(c(x,rev(x)),
        c(interactions[,1],rep(0,5)))

```

```

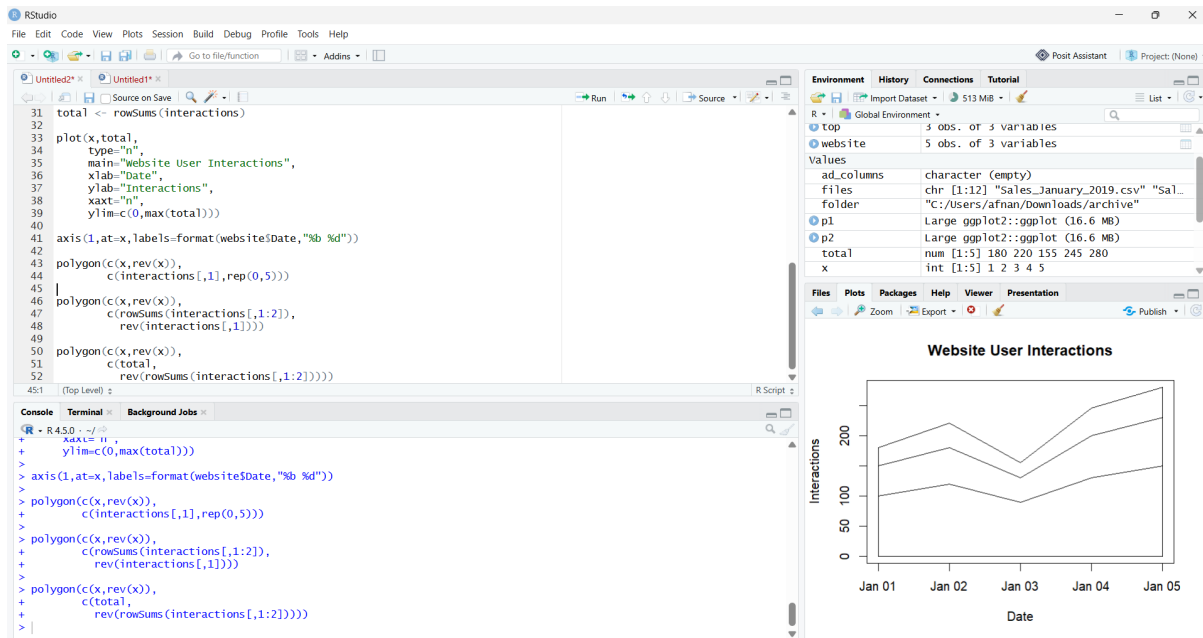
polygon(c(x,rev(x)),
        c(rowSums(interactions[,1:2]),
          rev(interactions[,1]))))

```

```

polygon(c(x,rev(x)),
c(total,
rev(rowSums(interactions[,1:2]))))

```



RESULT

Thus, the **Website Analytics** data was successfully analyzed using R programming. The **line chart**, **bar chart**, and **stacked area chart** were successfully created to visualize website traffic and user interaction patterns.

Note: The given Set 5 sheet provides Page Views and Click-through Rate, but it does not provide Likes, Shares, and Comments values. The interaction values used in the code are sample values solely for creating the required stacked area chart.